Lecture 10

Differential forms

Objectives. *A differential form supplies an integrand for curves, surfaces and higher-dimensional pieces of a manifold. At each point it assigns a signed multilinear measurement to tangent vectors. Forms pull back under every smooth map. Their exterior derivative extends the differential of a function and combines the Cartesian gradient, curl and divergence. We construct this derivative in coordinates, prove that it is independent of coordinates, and compute the sphere's area form.*

By the end of this lecture you will be able to:

- Evaluate forms with the determinant wedge convention.
- Compute exterior derivatives and pullbacks under smooth maps.
- Detect nonexactness of a closed 1-form by a nonzero loop integral.
- Derive both angular and stereographic expressions for sphere area.

10.1 Covectors and the cotangent bundle

Let M be a smooth n-manifold.

Definition 10.1 (Cotangent space)**.** The *cotangent space* at $p \in M$ is the dual vector space $T_p^*M = \mathrm{Hom}(T_pM, \mathbb{R})$. Its elements are *covectors*.

Definition 10.2 (Differential)**.** For a smooth function $f : M \to \mathbb{R}$, its *differential* is given by

$$(\mathrm{d}f)_p(v) = v(f) \qquad (v \in T_pM).$$

This is the differential of Lecture 8 with $T_{f(p)}\mathbb{R}$ identified with $\mathbb{R}$ by its standard coordinate. In coordinates $x^1, \dots, x^n$, the covectors $\mathrm{d}x^i$ are dual to ∂_j, since $\mathrm{d}x^i(\partial_j) = \delta^i_j$. Thus a covector has a unique expression $\alpha = \alpha_i \mathrm{d}x^i$, and $\alpha(v) = \alpha_i v^i$. In particular,

$$\mathrm{d}f = \frac{\partial f}{\partial x^i}\mathrm{d}x^i.$$

Definition 10.3 (Cotangent bundle)**.** The *cotangent bundle* is the disjoint union $T^*M = \coprod_{p\in M} T_p^*M$, with projection to p.

Its local coordinates are $(x^1, \dots, x^n, \alpha_1, \dots, \alpha_n)$. If $y = y(x)$ has Jacobian $J = Dy$, invariance of $\alpha(v)$ requires the coefficient row to change by $\alpha' = \alpha J^{-1}$. These are smooth invertible bundle transitions. They give the cotangent manifold by the same construction as sketched for the tangent bundle in Lecture 8. The Hausdorff and countable-basis arguments there apply with these transitions.

Definition 10.4 (Smooth 1-form)**.** A *smooth 1-form* is a smooth section of T^*M. Equivalently, its coefficients α_i are smooth in every chart.

The form $\mathrm{d}f$ is smooth because the partial derivatives of f are smooth.

Physical input. In classical mechanics momentum at a position is represented by a covector that pairs with velocity. Thus a point of T^*M records position and momentum. Lecture 11 uses these pairs as phase-space coordinates.

A covector measures a tangent vector, as the figure illustrates.

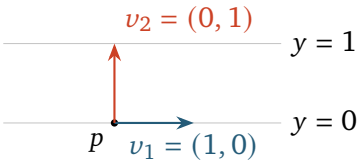


Figure 10.1: The horizontal lines are levels of the coordinate y. The covector $\mathrm{d}y$ gives 0 on the horizontal vector and 1 on the vertical vector.

For example, on $\mathbb{R}^2$ let $\alpha = 2x\mathrm{d}x + \mathrm{d}y$. At $(1, 0)$ it gives $\alpha(3\partial_x - \partial_y) = 6 - 1 = 5$. This form is the differential of $x^2 + y$.

10.2 Alternating tensors and the wedge product

A signed surface integrand measures a parallelogram in a tangent space. Linearity expresses addition and scaling in each edge. Equal edges span no area, so the value is zero. For a bilinear map, expansion on $(v + w, v + w)$ then shows that exchanging the edges reverses the sign. Alternating forms extend this rule to higher dimensions.

Definition 10.5 (Tensor field)**.** A *tensor field of type* (r, s) assigns at each point a multilinear real-valued map with r cotangent arguments and s tangent arguments. Its coordinate coefficients must be smooth. A tensor with $r = 0$ is *covariant*.

Definition 10.6 (Differential form)**.** A *k-form* is a covariant k-tensor field that is alternating: interchanging two arguments reverses its sign. We write $\Omega^k(M)$ for the vector space of smooth k-forms. A 0-form is a smooth function.

The degree k counts tangent arguments, whereas n is the dimension of M. Alternation implies that every k-form is zero if $k > \dim M$.

Recall the exterior powers from *Linear Algebra*, Lecture 12, the wedge-and-determinant example.

Proposition 10.7 (Exterior powers · *imported definition; evaluation proved*)**.** *For a finite-dimensional real vector space V, the space $\Lambda^k V^*$ consists of alternating k-linear forms on V. For covectors $\alpha_1, \dots, \alpha_k$, the determinant convention gives*

$$(\alpha_1 \wedge \cdots \wedge \alpha_k)(v_1, \dots, v_k) = \det\big(\alpha_i(v_j)\big)_{i,j=1}^{k}.$$

Proof of the evaluation formula. LA12 identifies covariant tensors with multilinear maps. Its antisymmetric tensors are therefore alternating maps. Evaluating its unnormalized antisymmetrization gives the sum $\sum_\pi \operatorname{sgn}(\pi) \prod_j \alpha_{\pi(j)}(v_j)$. The Leibniz determinant formula identifies this with the stated determinant. ∎

For two covectors the determinant formula reads

$$(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v).$$

In the plane, the form $\mathrm{d}x \wedge \mathrm{d}y$ measures signed area. The figure also shows the coordinate-plane areas measured by a 2-form in $\mathbb{R}^3$.

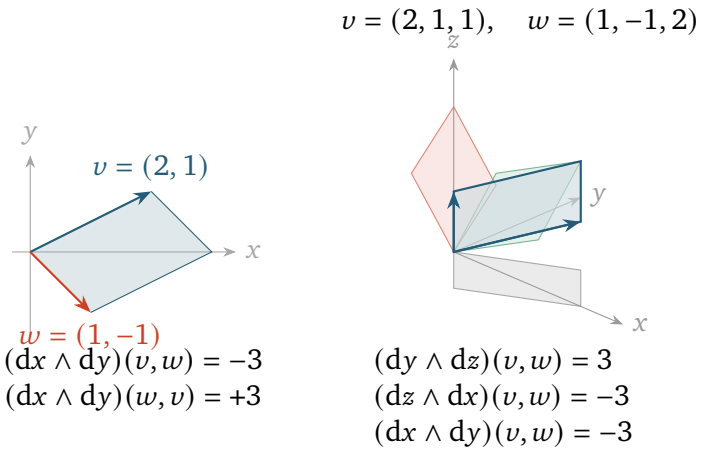


Figure 10.2: The ordered planar edges $(2, 1)$ and $(1, -1)$ give signed area -3; exchanging them gives $+3$. The projections of $(2, 1, 1)$ and $(1, -1, 2)$ have signed areas 3, -3 and -3 in the displayed coordinate orders.

Example 10.8 (A numerical 2-form)**.** On $\mathbb{R}^2$ take $\alpha = \mathrm{d}x + 2\mathrm{d}y$ and $\beta = 3\mathrm{d}x - \mathrm{d}y$. Bilinearity and alternation give

$$\alpha \wedge \beta = -\mathrm{d}x \wedge \mathrm{d}y + 6\mathrm{d}y \wedge \mathrm{d}x = -7\mathrm{d}x \wedge \mathrm{d}y.$$

For $v = (2, 1)$ and $w = (1, -1)$, the signed area is $(\mathrm{d}x \wedge \mathrm{d}y)(v, w) = 2(-1) - 1 \cdot 1 = -3$. The value of $\alpha \wedge \beta$ is therefore $(-7)(-3) = 21$.

For an increasing index tuple $I = (i_1 < \cdots < i_k)$, write $\mathrm{d}x^I = \mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}$. Every k-form has a unique expansion

$$\omega = \sum_I a_I \mathrm{d}x^I, \qquad a_I = \omega(\partial_{i_1}, \dots, \partial_{i_k}).$$

To prove this, the determinant gives $\mathrm{d}x^I(\partial_{j_1},\dots,\partial_{j_k}) = \delta^I_J$ for increasing tuples I and J. This forces the displayed coefficients. Conversely, subtracting the displayed sum from ω gives an alternating form vanishing on every increasing tuple of coordinate vectors. Alternation makes it zero on every ordered tuple, including repeated indices. Multilinearity then makes it zero on arbitrary vectors. This proves existence and uniqueness of the expansion. The sum runs over increasing tuples only, so it has no factorial. For a 2-form with component $\omega_{12} = c$ in two dimensions, alternation gives $\omega_{21} = -c$. The expression with all indices is

$$\frac{1}{2}\big(c\mathrm{d}x^1 \wedge \mathrm{d}x^2 - c\mathrm{d}x^2 \wedge \mathrm{d}x^1\big) = c\mathrm{d}x^1 \wedge \mathrm{d}x^2.$$

If instead every index ranges independently from 1 to n, set $\omega_{i_1\cdots i_k} = \omega(\partial_{i_1},\dots,\partial_{i_k})$. Then

$$\omega = \frac{1}{k!}\sum_{i_1,\dots,i_k} \omega_{i_1\cdots i_k}\mathrm{d}x^{i_1} \wedge \cdots \wedge \mathrm{d}x^{i_k}.$$

Each increasing tuple occurs $k!$ times with compensating signs in the component and basis wedge, which explains the factorial.

Definition 10.9 (Wedge product)**.** The *wedge product* of forms of degrees k and ℓ is the bilinear extension of concatenation of these basis wedges. A repeated index gives zero; otherwise sorting the concatenated indices gives its permutation sign.

For variable coefficients the sorting rule gives

$$(x\mathrm{d}x + y\mathrm{d}y) \wedge (z\mathrm{d}x + x\mathrm{d}y) = (x^2 - yz)\mathrm{d}x \wedge \mathrm{d}y.$$

For a 2-form α and a 1-form β, the intrinsic evaluation is

$$(\alpha \wedge \beta)(u,v,w) = \alpha(u,v)\beta(w) - \alpha(u,w)\beta(v) + \alpha(v,w)\beta(u).$$

In the full permutation sum, each term appears twice because interchanging the two arguments of α reverses both signs. Division by 2! leaves these three terms. The general formula has the same cancellation within each block. This definition agrees with the determinant convention. Intrinsically it is

$$(\alpha \wedge \beta)(v_1,\dots,v_{k+\ell}) = \frac{1}{k!\ell!}\sum_{\pi\in S_{k+\ell}} \operatorname{sgn}(\pi)\alpha(v_{\pi(1)},\dots,v_{\pi(k)})\beta(v_{\pi(k+1)},\dots,v_{\pi(k+\ell)}).$$

Indeed, grouping the permutations by which k arguments enter α leaves $k!\ell!$ identical contributions within each group. By bilinearity take $\alpha = \mathrm{d}x^I$ and $\beta = \mathrm{d}x^J$. Both the permutation expression and the sorted basis wedge are alternating in their arguments, so the preceding basis test reduces their comparison to increasing tuples of coordinate vectors. Only shuffles placing the first index block in its matching slots contribute. They give its sorting sign; a repeated index gives zero. Thus the formula equals the sorting rule. The intrinsic formula proves that wedge is independent of coordinates and has smooth coefficients. Concatenating three index lists proves associativity: both parenthesizations have the same sorting sign, or both vanish because an index repeats. Moving a block of k indices past a block of ℓ indices makes $k\ell$ swaps. Bilinearity therefore gives graded commutativity:

$$\alpha \wedge \beta = (-1)^{k\ell}\beta \wedge \alpha.$$

As a tensor, the determinant wedge is $\alpha\wedge\beta = \alpha\otimes\beta - \beta\otimes\alpha$. The normalized fermionic tensor of *QM-I*, Lecture 14, instead uses $(u\otimes v - v\otimes u)/\sqrt{2}$ for orthonormal one-particle states. For k

orthonormal factors its normalization is $1/\sqrt{k!}$ times the unnormalized antisymmetrization.[1] Our convention gives $(\mathrm{d}x \wedge \mathrm{d}y)(\partial_x, \partial_y) = 1$. This value fixes the normalization used by the Hodge operator in Lecture 15.

Self-check. State in one sentence what a differential k-form is and why it is introduced. Explain why repeated tangent arguments give zero.

Answer. A differential k-form is a smooth alternating multilinear measurement of k tangent vectors at each point, introduced as an integrand for signed k-dimensional integration. Exchanging two equal arguments leaves the value unchanged and negates it, so the value is zero.

10.3 The exterior derivative

Definition 10.10 (Exterior derivative)**.** The *exterior derivative* raises form degree by one. On a coordinate chart define it by

$$\mathrm{d}\omega = \sum_I \sum_j (\partial_j a_I)\mathrm{d}x^j \wedge \mathrm{d}x^I \quad \text{when} \quad \omega = \sum_I a_I \mathrm{d}x^I.$$

On functions this agrees with the differential already defined.

Definition 10.11 (Line integral)**.** For a smooth 1-form α on M and a smooth path $\gamma\colon [a,b] \to M$, define the line integral by

$$\int_\gamma \alpha = \int_a^b \alpha_{\gamma(t)}(\gamma'(t))\,\mathrm{d}t.$$

Here a smooth path on $[a,b]$ means the restriction of a smooth curve on an open interval containing $[a,b]$. For a 1-form $\alpha = a\mathrm{d}x + b\mathrm{d}y$ on an open set in $\mathbb{R}^2$, exterior differentiation gives

$$\mathrm{d}\alpha = (\partial_x b - \partial_y a)\mathrm{d}x \wedge \mathrm{d}y.$$

This coefficient measures circulation per unit area. To see this directly, let C_h be the line integral around a counter-clockwise square of side $h > 0$ based at $p = (x_0, y_0)$, with the square contained in the open set. Pairing the horizontal edges and then the vertical edges gives

$$\begin{aligned} C_h &= \int_0^h \big[a(x_0 + t, y_0) - a(x_0 + t, y_0 + h)\big]\,\mathrm{d}t \\ &\quad + \int_0^h \big[b(x_0 + h, y_0 + t) - b(x_0, y_0 + t)\big]\,\mathrm{d}t \\ &= h^2(\partial_x b - \partial_y a)_p + o(h^2). \end{aligned}$$

The fundamental theorem of calculus expresses each bracket as an integral of a first derivative. Continuity of these derivatives gives the last line as $h \to 0^+$. Its leading term is $\mathrm{d}\alpha_p(h\partial_x, h\partial_y)$. Lecture 12 proves the corresponding equality between boundary and surface integrals in Stokes' theorem.

For $\alpha = -y\mathrm{d}x + x\mathrm{d}y$, the derivative is $2\mathrm{d}x \wedge \mathrm{d}y$. The figure records the four edge integrals on the square $[0,h]^2$.

[1] QM-I, Lecture 14, Many fermions: Slater determinants, uses normalized tensor states. The present wedge acts on covectors and uses determinant normalization.

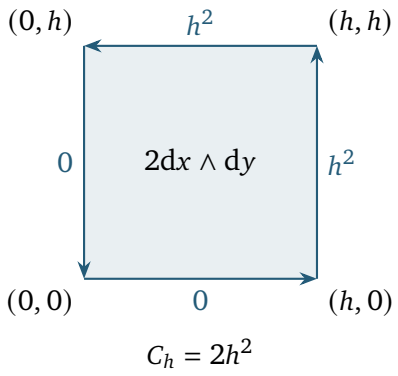


Figure 10.3: For $\alpha = -y\mathrm{d}x + x\mathrm{d}y$, the counter-clockwise edge integrals are 0, h^2, h^2 and 0. Their sum is $2h^2$, in agreement with the exterior derivative.

Theorem 10.12 (Exterior derivative · *sketched; existence and identities proved*)**.** *The coordinate formula defines a coordinate-independent real-linear map* $\mathrm{d}\colon \Omega^k(M) \to \Omega^{k+1}(M)$. *It satisfies* $\mathrm{d}^2 = 0$ *and, for* $\alpha \in \Omega^k(M)$,

$$\mathrm{d}(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^k \alpha \wedge \mathrm{d}\beta.$$

These properties and its value on functions determine it uniquely.

Proof of existence and identities. Work first in one chart. The formula $\mathrm{d}(b\mathrm{d}x^J) = \mathrm{d}b \wedge \mathrm{d}x^J$ also holds for a tuple J that is not increasing: reordering changes both sides by the same sign, and a repeated index makes both sides zero. Applying d twice gives terms $(\partial_r \partial_j a_I)\mathrm{d}x^r \wedge \mathrm{d}x^j \wedge \mathrm{d}x^I$. Terms with $r = j$ vanish. Each remaining term cancels its partner with j and r exchanged, because mixed derivatives agree and wedge is antisymmetric. Thus $\mathrm{d}^2 = 0$.

For the product rule, expand into terms $\alpha = a\mathrm{d}x^I$ and $\beta = b\mathrm{d}x^J$, where $|I| = k$. The ordinary product rule differentiates ab into $b\mathrm{d}a + a\mathrm{d}b$. The first contribution is $\mathrm{d}\alpha \wedge \beta$. In the second, moving $\mathrm{d}b$ past the k factors of $\mathrm{d}x^I$ gives $(-1)^k \alpha \wedge \mathrm{d}\beta$. Summing proves the identity.

Now let $y = y(x)$ be another chart. The chain rule gives $\mathrm{d}y^a = (\partial_i y^a)\mathrm{d}x^i$. The operator defined in x coordinates annihilates $\mathrm{d}y^a$ by the same mixed-derivative cancellation. For a coefficient function b, the chain rule also identifies its x and y differentials. Apply the product rule just proved to a monomial $b\mathrm{d}y^{i_1} \wedge \cdots \wedge \mathrm{d}y^{i_k}$. Only the differentiated coefficient survives, so the result is precisely the y coordinate formula. Such monomials span every form. Thus the definitions agree on chart overlaps and give a global smooth form. ∎

The uniqueness assertion is sketched in the optional §10.8.

For example, $\mathrm{d}(x\mathrm{d}y) = \mathrm{d}x \wedge \mathrm{d}y$, while $\mathrm{d}(x\mathrm{d}x) = \mathrm{d}x \wedge \mathrm{d}x = 0$.

10.4 Pullback under a smooth map

Let $f\colon M \to N$ be any smooth map.

Definition 10.13 (Pullback)**.** For $\omega \in \Omega^k(N)$ its *pullback* is given by

$$(f^*\omega)_p(v_1, \ldots, v_k) = \omega_{f(p)}(\mathrm{d}f_p v_1, \ldots, \mathrm{d}f_p v_k).$$

For functions, set $f^*h = h \circ f$.

The differential sends each argument to the required target tangent space. No inverse of f or of its differential is needed.

Theorem 10.14 (Pullback calculus). *Pullback takes smooth forms to smooth forms and satisfies*

$$f^*(\alpha \wedge \beta) = f^*\alpha \wedge f^*\beta, \qquad \mathrm{d}(f^*\alpha) = f^*(\mathrm{d}\alpha).$$

For smooth $g\colon N \to P$ it also satisfies $(g \circ f)^* = f^*g^*$ and $\mathrm{id}^* = \mathrm{id}$.

Proof. Substituting $\mathrm{d}f_p v_j$ in the permutation formula proves wedge compatibility. The chain rule gives $f^*\mathrm{d}h = \mathrm{d}(h \circ f)$. In local target coordinates y, write $f^a = y^a \circ f$. The pullback of $\alpha = \sum_I a_I \mathrm{d}y^I$ is

$$f^*\alpha = \sum_I (a_I \circ f)\mathrm{d}f^{i_1} \wedge \cdots \wedge \mathrm{d}f^{i_k}.$$

Its coefficients are smooth. Differentiating this expression leaves only $\mathrm{d}(a_I \circ f)$, since $\mathrm{d}^2 f^a = 0$. Pulling back $\mathrm{d}\alpha$ gives the same expression by the chain rule on coefficient functions. This proves commutation with d without using the uniqueness sketch. Finally, substituting $\mathrm{d}(g \circ f) = \mathrm{d}g \circ \mathrm{d}f$ into the definition proves the composition rule, and identity differentials prove the identity rule. ∎

For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, pullback gives $f^*(\mathrm{d}y) = 2x\mathrm{d}x$, including at $x = 0$. Lecture 9 showed that ∂_x does not descend under this same map. Forms pull back even when a map loses rank or identifies distinct points.

Proposition 10.15 (Determinant rule for top-form pullback). *Let $U, V \subseteq \mathbb{R}^n$ be open and let $f\colon U \to V$ be smooth. For a smooth function $a\colon V \to \mathbb{R}$,*

$$f^*(a\mathrm{d}y^1 \wedge \cdots \wedge \mathrm{d}y^n) = (a \circ f)\det(Df)\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^n.$$

Proof. Both sides are top forms on U. Evaluating on $(\partial_1, \ldots, \partial_n)$ gives $(a \circ f)\det(Df)$ on the left by the determinant wedge formula. The right side has this same coefficient, so the forms agree. ∎

We compute a pullback by substituting the component functions of the map and replacing each target differential by the differential of that component.

Example 10.16 (Polar pullback). Let $f(r, \theta) = (r\cos\theta, r\sin\theta)$, where $r > 0$ and θ belongs to an open interval of length less than 2π. The component differentials are

$$f^*\mathrm{d}x = \cos\theta\,\mathrm{d}r - r\sin\theta\,\mathrm{d}\theta, \qquad f^*\mathrm{d}y = \sin\theta\,\mathrm{d}r + r\cos\theta\,\mathrm{d}\theta.$$

Their wedge is

$$f^*(\mathrm{d}x \wedge \mathrm{d}y) = r(\cos^2\theta + \sin^2\theta)\mathrm{d}r \wedge \mathrm{d}\theta = r\mathrm{d}r \wedge \mathrm{d}\theta.$$

The determinant rule identifies the coefficient r as the Jacobian determinant.

Self-check. For $f\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(u, v) = (u^2 - v, uv)$ and the target 1-form $\alpha = x\mathrm{d}y$, compute $f^*\alpha$ and $\mathrm{d}(f^*\alpha)$. Compute $f^*(\mathrm{d}\alpha)$ independently and compare.

Answer. Substitution gives

$$f^*\alpha = (u^2 - v)(v\mathrm{d}u + u\mathrm{d}v).$$

The $\mathrm{d}u \wedge \mathrm{d}v$ coefficient of its derivative is $\partial_u[u(u^2 - v)] - \partial_v[v(u^2 - v)] = 2u^2 + v$. On the target, $\mathrm{d}\alpha = \mathrm{d}x \wedge \mathrm{d}y$. The determinant of the derivative of f is $\det\left(\begin{smallmatrix} 2u & -1 \\ v & u \end{smallmatrix}\right) = 2u^2 + v$. Thus both computations give $(2u^2 + v)\mathrm{d}u \wedge \mathrm{d}v$.

10.5 Gradient, curl and divergence

Fix Cartesian coordinates on Euclidean $\mathbb{R}^3$. Identify a vector (A_1, A_2, A_3) with the 1-form $\alpha = A_1\mathrm{d}x + A_2\mathrm{d}y + A_3\mathrm{d}z$. Identify (B_1, B_2, B_3) with the 2-form $\beta = B_1\mathrm{d}y\wedge\mathrm{d}z + B_2\mathrm{d}z\wedge\mathrm{d}x + B_3\mathrm{d}x\wedge\mathrm{d}y$. Identify a scalar g with the 3-form $g\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z$. These are specified coordinate identifications. The 1-form evaluates on one tangent vector by the Euclidean dot product. The 2-form evaluates on an ordered pair by the dot product with their cross product. The second identification also uses the orientation (x, y, z).

Physical input. For a force field in Euclidean space, work along a path is the line integral of its dot product with velocity. For a fluid velocity field, volume flux through an oriented surface is the surface integral of its normal component.

The 1-form therefore represents work along curves. The 2-form represents flux through surfaces. The differential $\mathrm{d}f$ exists without a metric. Its associated gradient vector uses the Euclidean dot product through $\mathrm{d}f(v) = \operatorname{grad} f \cdot v$.

Proposition 10.17 (Vector calculus from forms)**.** *Under these identifications, exterior differentiation gives gradient, curl and divergence in successive degrees.*

Proof. The defining formula gives the following coefficients:

$$\begin{aligned}
\mathrm{d}f &= (\partial_x f)\mathrm{d}x + (\partial_y f)\mathrm{d}y + (\partial_z f)\mathrm{d}z,\\
\mathrm{d}\alpha &= (\partial_y A_3 - \partial_z A_2)\mathrm{d}y\wedge\mathrm{d}z + (\partial_z A_1 - \partial_x A_3)\mathrm{d}z\wedge\mathrm{d}x\\
&\quad + (\partial_x A_2 - \partial_y A_1)\mathrm{d}x\wedge\mathrm{d}y,\\
\mathrm{d}\beta &= (\partial_x B_1 + \partial_y B_2 + \partial_z B_3)\mathrm{d}x\wedge\mathrm{d}y\wedge\mathrm{d}z.
\end{aligned}$$

For example, the $\mathrm{d}y\wedge\mathrm{d}z$ coefficient receives $\partial_y A_3$ from $\mathrm{d}(A_3\mathrm{d}z)$ and $-\partial_z A_2$ from $\mathrm{d}(A_2\mathrm{d}y)$. Cyclic permutation gives the other two coefficients. For $\mathrm{d}\beta$, repeated factors vanish and each surviving cyclic order has positive sign. The displayed coefficients are the Cartesian definitions of gradient, curl and divergence. ∎

Applying $\mathrm{d}^2 = 0$ in degrees zero and one respectively gives

$$\operatorname{curl}\operatorname{grad} f = 0, \qquad \operatorname{div}\operatorname{curl} A = 0.$$

10.6 Closed and exact forms

Definition 10.18 (Closed form)**.** A form ω is *closed* if $\mathrm{d}\omega = 0$.

Definition 10.19 (Exact form)**.** A k-form with $k \geq 1$ is *exact* if it equals $\mathrm{d}\eta$ for some $(k-1)$-form η.

Every exact form is closed by $\mathrm{d}^2 = 0$. For a smooth 1-form α on M and a smooth path $\gamma\colon [a, b] \to M$, the pullback definition gives $\gamma^*\alpha = \alpha_{\gamma(t)}(\gamma'(t))\mathrm{d}t$, so this is also the integral of the pullback's coefficient. If $\alpha = \mathrm{d}f$, the chain rule and the fundamental theorem of calculus give $\int_\gamma \mathrm{d}f = f(\gamma(b)) - f(\gamma(a))$.

Proposition 10.20 (A closed form that is not exact). *On* $\mathbb{R}^2 \setminus \{0\}$ *the angular form*

$$\omega_{\text{ang}} = \frac{x\mathrm{d}y - y\mathrm{d}x}{x^2 + y^2}$$

is closed but not exact.

Proof. Its exterior derivative has $\mathrm{d}x \wedge \mathrm{d}y$ coefficient

$$\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2} = \frac{2}{x^2+y^2} - \frac{2(x^2+y^2)}{(x^2+y^2)^2} = 0.$$

For $\gamma(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$, substitution gives $\gamma^*\omega_{\text{ang}} = \mathrm{d}t$. Its integral is 2π. If it were exact, the endpoint formula would give zero on this closed path. This contradiction proves nonexactness. ∎

For a smooth loop $\gamma(t) = r(t)(\cos\phi(t), \sin\phi(t))$, where $r(t) > 0$, differentiation gives $x\dot{y} - y\dot{x} = r^2\dot{\phi}$. Thus $\gamma^*\omega_{\text{ang}} = \dot{\phi}\mathrm{d}t$. Lecture 3 lifts $\gamma/|\gamma|$ in turns, as drawn in its continuous-phase figure. Multiplying that lift by 2π gives ϕ. The angle ϕ is smooth because it locally differs from a smooth angle branch along γ by a constant multiple of 2π. The integral divided by 2π is therefore the winding number. The same computation with variable r and ϕ gives $\omega_{\text{ang}} = \mathrm{d}\phi$ on an angle chart. There is no global smooth angle function on the punctured plane. On the polar chart of §10.4, substitution gives $f^*\omega_{\text{ang}} = \mathrm{d}\theta$: the numerator pulls back to $r^2\mathrm{d}\theta$ and the denominator to r^2. Thus a counter-clockwise circle of any radius $R > 0$ has integral 2π. The angle branch $\phi_1 \in (-\pi, \pi)$ is defined off the nonpositive x-axis. The branch $\phi_2 \in (0, 2\pi)$ is defined off the nonnegative x-axis. On the upper half-plane the branches satisfy $\phi_2 = \phi_1$. On the lower half-plane they satisfy $\phi_2 = \phi_1 + 2\pi$. Their differentials agree on both components of the overlap. The continuous phase of Lecture 3 records a full turn even when the planar endpoints agree. The two angle charts in the figure show why their local primitives do not give a single global function.

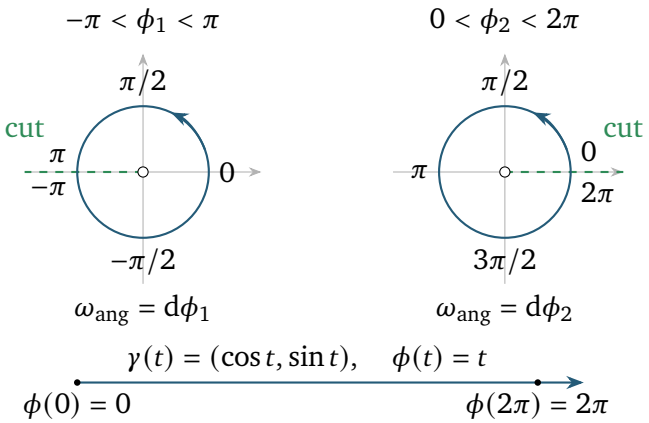


Figure 10.4: The angle branches have different radial cuts and differ by a constant on each component of the overlap. The continuous angle along a positive loop changes by 2π although the planar endpoints agree.

10.7 The area form of the sphere

The vector-calculus dictionary assigns the ambient 2-form β to the radial field (x, y, z). It measures the field's flux through a tangent parallelogram. On the unit sphere this field

is the outward unit normal, so its flux measures signed area. The inclusion $i\colon S^2 \to \mathbb{R}^3$ pulls the ambient form $\beta = x\,dy\wedge dz + y\,dz\wedge dx + z\,dx\wedge dy$ back to $\omega_{S^2} = i^*\beta$. At $p \in S^2$, for $v, w \in T_pS^2$, it evaluates as

$$(\omega_{S^2})_p(v,w) = \det[p\ v\ w] = p\cdot(v\times w).$$

Expanding the determinant along its first column gives

$$x(v_2w_3 - v_3w_2) + y(v_3w_1 - v_1w_3) + z(v_1w_2 - v_2w_1),$$

which is the evaluation of β at $p = (x,y,z)$. This is signed area measured with the outward unit normal p.

Example 10.21 (Angular area). For angular parameters put

$$s(\theta,\phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta),$$

with $0 < \theta < \pi$ and ϕ in an open interval of length less than 2π. Its coordinate tangents are

$$s_\theta = (\cos\theta\cos\phi, \cos\theta\sin\phi, -\sin\theta), \qquad s_\phi = (-\sin\theta\sin\phi, \sin\theta\cos\phi, 0).$$

Their cross product is $s_\theta \times s_\phi = \sin\theta\, s$, so

$$s^*\omega_{S^2} = \sin\theta\, d\theta\wedge d\phi.$$

For the inverse stereographic map from the north pole, set $\rho = 1 + u^2 + v^2$ and $\sigma(u,v) = (2u, 2v, u^2+v^2-1)/\rho$. At $(u,v) = (0,0)$ the point is $p = (0,0,-1)$, while the tangent columns are $\sigma_u = 2e_1$ and $\sigma_v = 2e_2$. Their cross product is $4e_3$, opposite to the outward normal p. The figure shows both tangents in the plane $z = -1$.

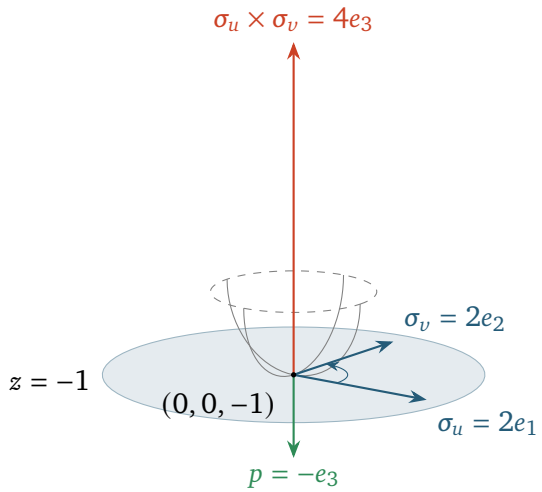


Figure 10.5: The shaded ellipse lies in the tangent plane $z = -1$ at the south pole. The ordered tangents have cross product $+4e_3$, opposite to the outward normal $p = -e_3$.

Example 10.22 (Stereographic area)**.** The derivative matrix of σ is

$$(\sigma_u\ \sigma_v) = \frac{1}{\rho^2}\begin{pmatrix} 2(\rho - 2u^2) & -4uv \\ -4uv & 2(\rho - 2v^2) \\ 4u & 4v \end{pmatrix}.$$

The cross product of these columns is

$$\sigma_u \times \sigma_v = \frac{(-8u\rho, -8v\rho, 4\rho(1 - u^2 - v^2))}{\rho^4} = -\frac{4\sigma}{\rho^2}.$$

Since $\sigma \cdot \sigma = 1$, the pullback is

$$\sigma^*\omega_{S^2} = -\frac{4}{(1 + u^2 + v^2)^2}\mathrm{d}u \wedge \mathrm{d}v.$$

The south-pole sign test agrees with the negative coefficient. Lecture 12 formalizes this comparison as orientation reversal.

Self-check. Compute $\mathrm{d}(x\mathrm{d}y \wedge \mathrm{d}z)$ on $\mathbb{R}^3$ and its pullback to S^2.

Answer. The derivative is $\mathrm{d}x \wedge \mathrm{d}y \wedge \mathrm{d}z$. Its pullback is zero: an alternating trilinear form vanishes on the two-dimensional tangent space of the sphere.

10.8 Uniqueness of the exterior derivative *(optional)*

The pullback, vector-calculus and area calculations do not depend on this uniqueness argument. Later results on integration and Cartan's formula also do not depend on it.

Sketch of uniqueness. Let d' have the properties in Theorem 10.12. A smooth *cutoff function* equals 1 near a chosen point and 0 outside a chosen chart neighbourhood. Such functions are constructed in a smaller coordinate ball using $e^{-1/t}$ for $t > 0$, extended by zero for $t \le 0$. If α vanishes near p, choose a cutoff χ supported where $\alpha = 0$. Since $\chi\alpha = 0$, the product rule at p gives $(\mathrm{d}'\alpha)_p = 0$. Thus d' depends only on the form near p. Use a cutoff to extend local coefficient and coordinate functions smoothly to M, without changing them near p. The identity $\mathrm{d}'\mathrm{d}f = (\mathrm{d}')^2 f = 0$ for these extensions and the product rule force $\mathrm{d}'(\sum_I a_I \mathrm{d}x^I) = \sum_I \mathrm{d}a_I \wedge \mathrm{d}x^I$ at p. This proves uniqueness, with the cutoff construction left at sketch level. See Tu, §19.4. ∎

Summary. Forms alternate in their tangent arguments and use the determinant wedge. The exterior derivative is independent of coordinates, squares to zero and satisfies the graded product rule. Pullback exists under every smooth map and commutes with this derivative. A closed form need not be exact, as the angular form's nonzero loop integral proves. The sphere's outward area has negative coefficient in the north-pole stereographic coordinates. Lecture 11 differentiates forms along flows.

Reading. Tu, *An Introduction to Manifolds*, §§17–19; *Linear Algebra*, Lecture 12, exterior powers and the determinant wedge.